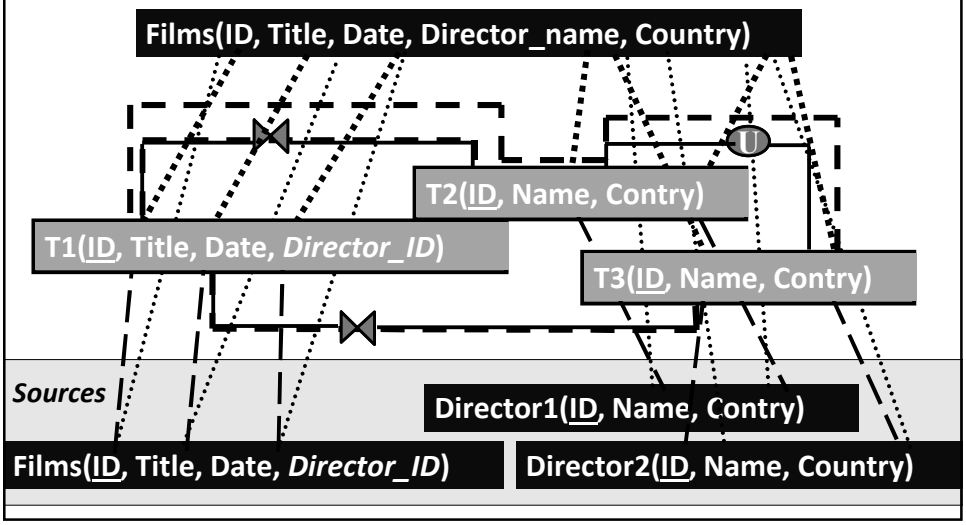
Generating mediation queries

Vu Tuyet Trinh
SoICT - HUST

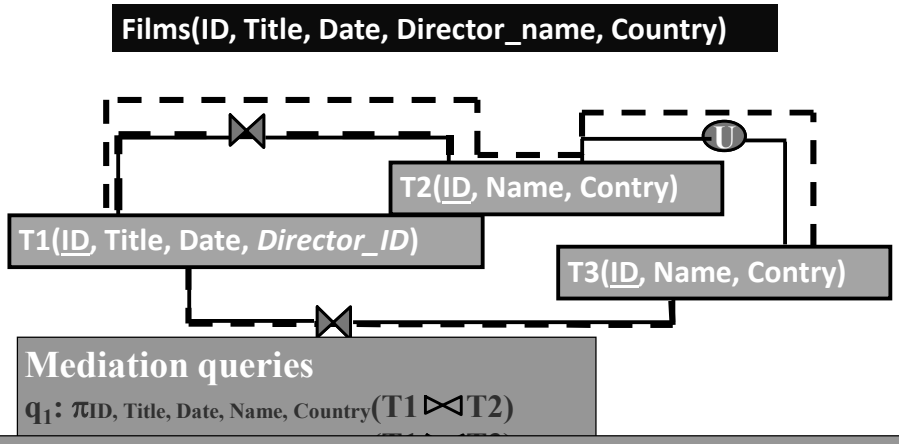
1

A Semi-automatic approach



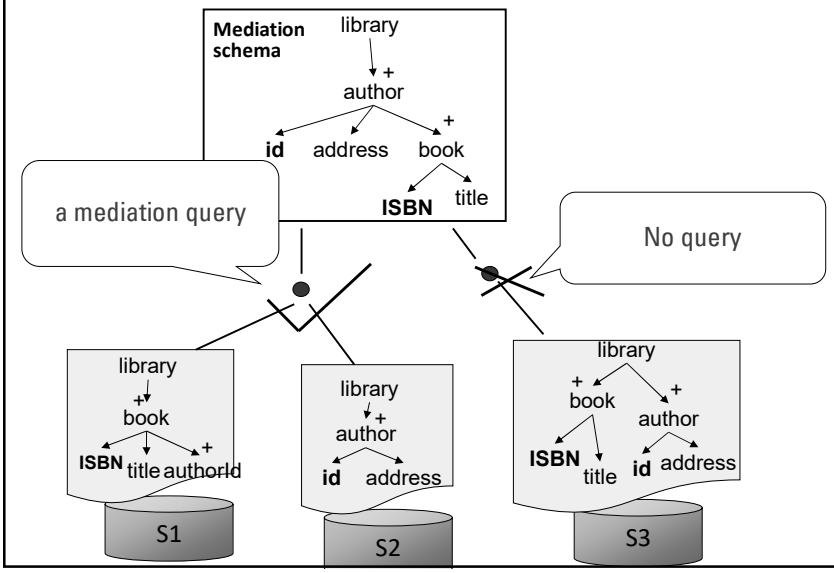
2

A Semi-automatic approach (2)



3

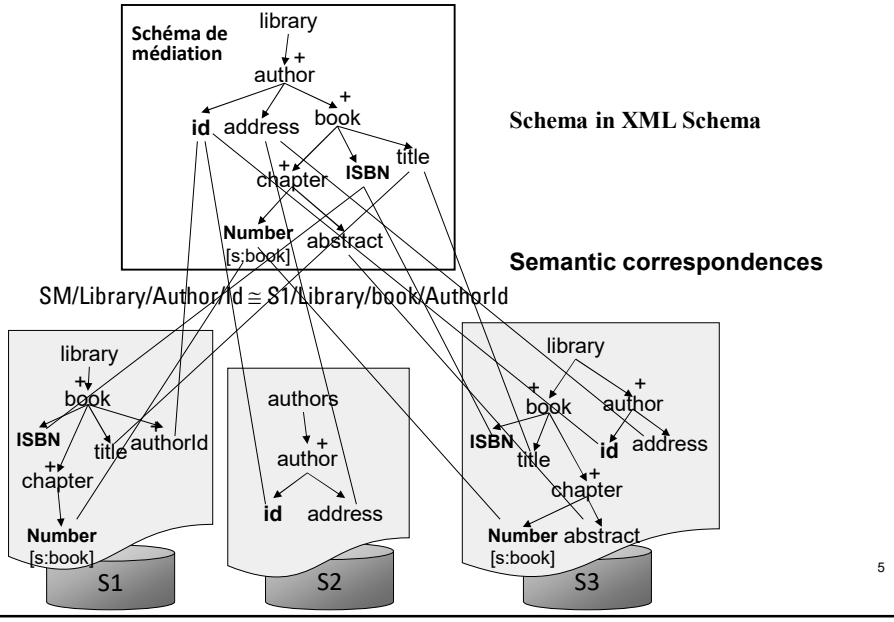
Extension for semi/un-structure data



4

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Assumption

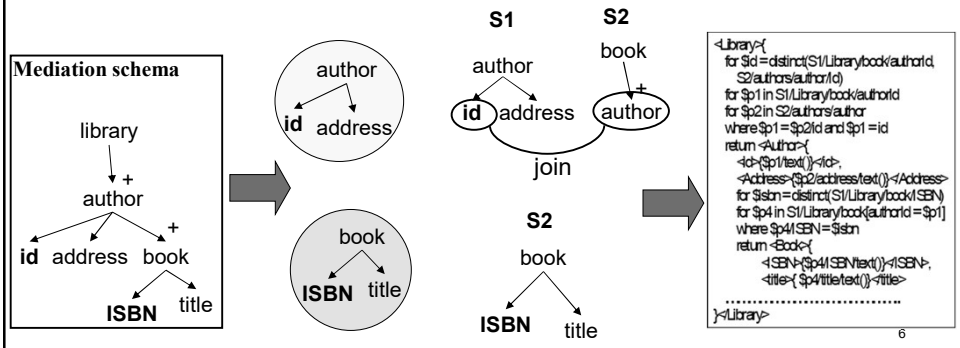


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Generating mediation queries

- Principle
  - Decomposing the mediation schema into sub-trees
  - Looking for partial mappings
  - Identifying mediation queries : combining partial mappings

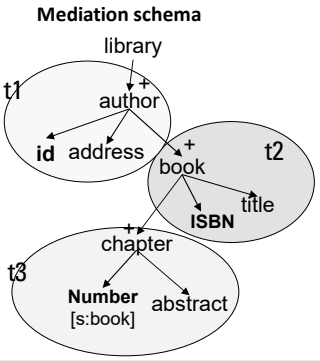


6

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Decomposing

- **Decomposing the mediation schema into sub-trees having**
  - Root: a multi-value node or the root of mediation schema
  - Node: mono-value
  - Existing at least a text node in sub-tree

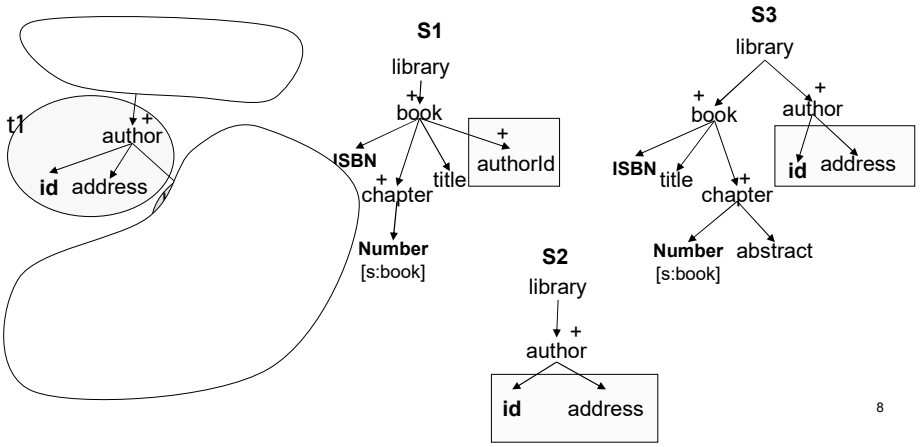


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Partial mappings

(1) For each sub-tree, identifying correspondent parties in sources



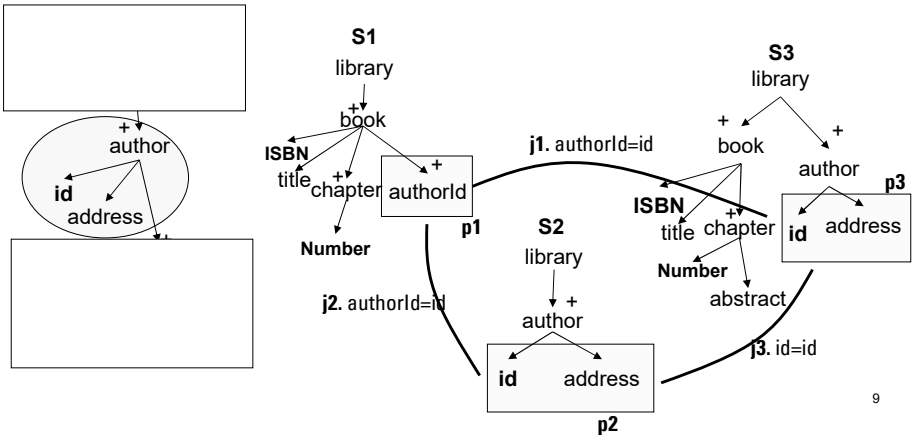
8

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Partial mapping (2)

(2) Identifying possible join operations between sources

- semantic correspondences

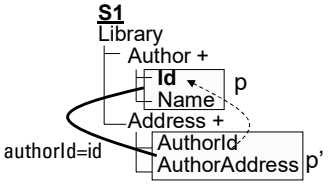


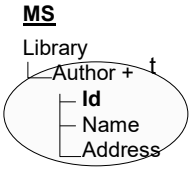
9

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Partial mapping (3)

- Identifying possible inter-join operation
  - Key & reference key



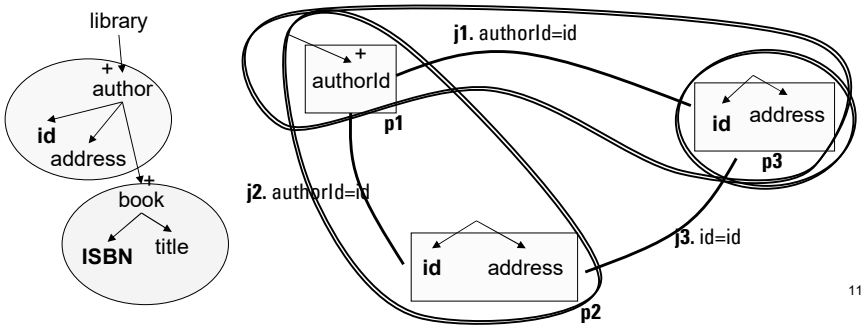


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Partial mapping (4)

- Graph of joins join :
  - Node ~ a source
  - axe ~ a join
- Partial mapping: set of sub-graph of joins



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Generating mediation queries

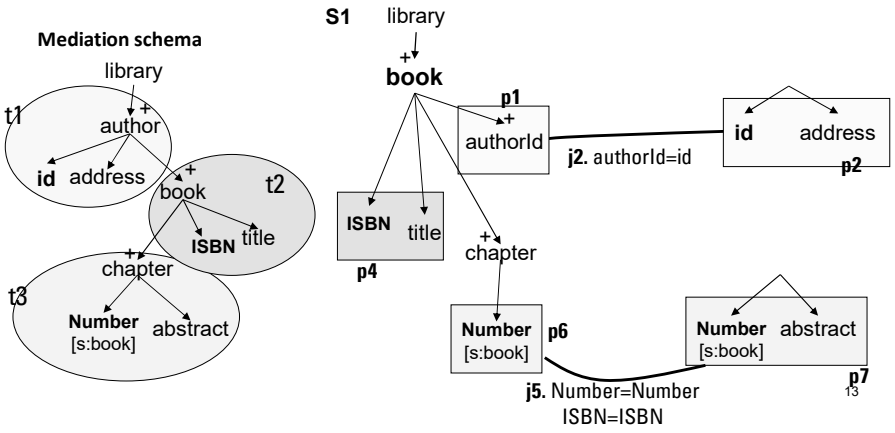
- **A mediation query ~ a combine of partial mappings**
  - A partial mapping for each sub-tree
  - Having some mediation queries (with different semantics)
- **Union of 2 mediation queries ~ a mediation query**

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Generating ... (2)

- **Verifying parent-child relations in sub-trees**



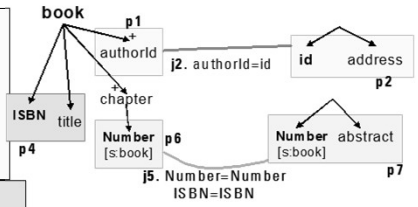
13

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Generating mediation queries

- Requête XQuery

```
<Library>{
for $id = distinct(S1/Library/book/authorId,
          S2/authors/author/Id)
for $p1 in S1/Library/book/authorId
for $p2 in S2/authors/author
where $p1 = $p2/id and $p1 = id
return <Author>{
       <id>{$p1/text()}</id>,
       <Address>{$p2/address/text()}</Address>
       for $isbn = distinct(S1/Library/book/ISBN)
       for $p4 in S1/Library/book[authorId = $p1]
       where $p4/ISBN = $isbn
       return <Book>{
              <ISBN>{$p4/ISBN/text()}</ISBN>,
              <title>{ $p4/title/text()}</title>
              for $num = distinct(S1/Library/book/Chapter/Number,
S3/Library/book/Chapter/Number)
              for $p6 in S1/Library/book[ISBN = $p4/ISBN]/Chapter
              for $p7 in S3/Library/book/Chapter
              where $p6/Number = $p7/Number and $p6/Number = $num
               and $p6/PARENT::book/ISBN= $p7/PARENT::book/ISBN
              return <Chapter>{
               ……..
  }</Library>
```



14

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